



TEHNIČKO VELEUČILIŠTE
U ZAGREBU UNIVERSITY OF
APPLIED SCIENCES ZAGREB



**Stručni prijediplomski
studij Elektrotehnika**

3. semestar

**Komunikacijska i
računalna tehnika**

Ak.g. 2025./2026.

Signali i procesi

Fourierova transformacija

Kalendar nastave po tjednima (ver.2)



#	Ponedjeljak	Utorak	Srijeda	Četvrtak	Petak
1	06.10.2025	07.10.2025	08.10.2025	09.10.2025	10.10.2025
2	13.10.2025	14.10.2025	15.10.2025	16.10.2025	17.10.2025
3	20.10.2025	21.10.2025	22.10.2025	23.10.2025	24.10.2025
4	27.10.2025	28.10.2025	29.10.2025	30.10.2025	31.10.2025
5	03.11.2025	04.11.2025	05.11.2025	06.11.2025	07.11.2025
6	10.11.2025	11.11.2025	12.11.2025	13.11.2025	14.11.2025
7	17.11.2025	18.11.2025	19.11.2025	20.11.2025	21.11.2025
8	24.11.2025	25.11.2025	26.11.2025	27.11.2025	28.11.2025
9	01.12.2025	02.12.2025	03.12.2025	04.12.2025	05.12.2025
10	08.12.2025	09.12.2025	10.12.2025	11.12.2025	12.12.2025
11	15.12.2025	16.12.2025	17.12.2025	18.12.2025	19.12.2025
	22.12.2025	23.12.2025	24.12.2025	25.12.2025	26.12.2025
	29.12.2025	30.12.2025	31.12.2025	01.01.2026	02.01.2026
12	05.01.2026	06.01.2026	07.01.2026	08.01.2026	09.01.2026
13	12.01.2026	13.01.2026	14.01.2026	15.01.2026	16.01.2026
14	19.01.2026	20.01.2026	21.01.2026	22.01.2026	23.01.2026
15	26.01.2026	27.01.2026	28.01.2026	29.01.2026	30.01.2026
16	02.02.2026	03.02.2026	04.02.2026	05.02.2026	06.02.2026

HRO-CIGRE

Samostalan rad

SiP 1. KZ

LV1

LV2

LV3

LV4

LV5

SiP 2. KZ

SiP LV G1
9:15 - 11:30

SiP LV G2
16:15 - 18:30

SiP PRED/AV
9:15 - 13:00

SiP LV G3
13:45 - 16:00

SiP LV G4
16:15 - 18:30

Prezentacija je samo podsjetnik o onome što je obuhvaćeno predavanjem i njezin sadržaj nije dovoljan za učenje!

Obavezna literatura – skripta:

- P. Valožić: Signali i procesi, skripta ,TVZ, 2015
Dostupno preko mojTVZ web stranice predmeta kao 8 pdf dokumenata:

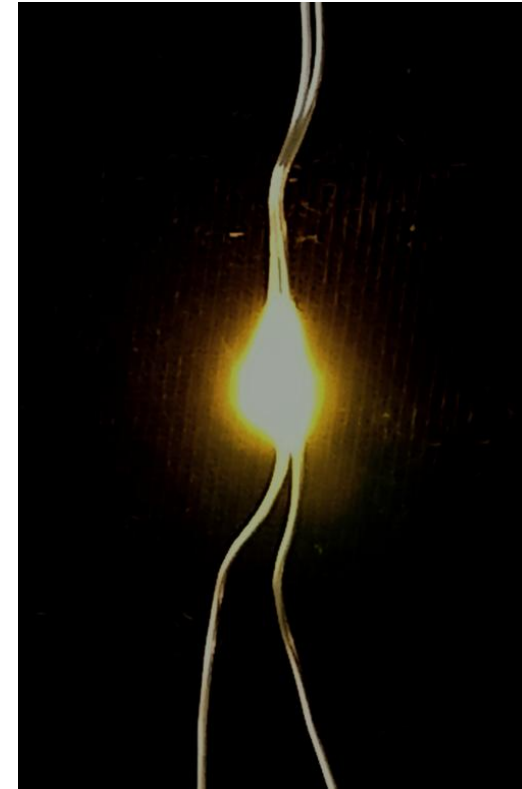
Udžbenici:

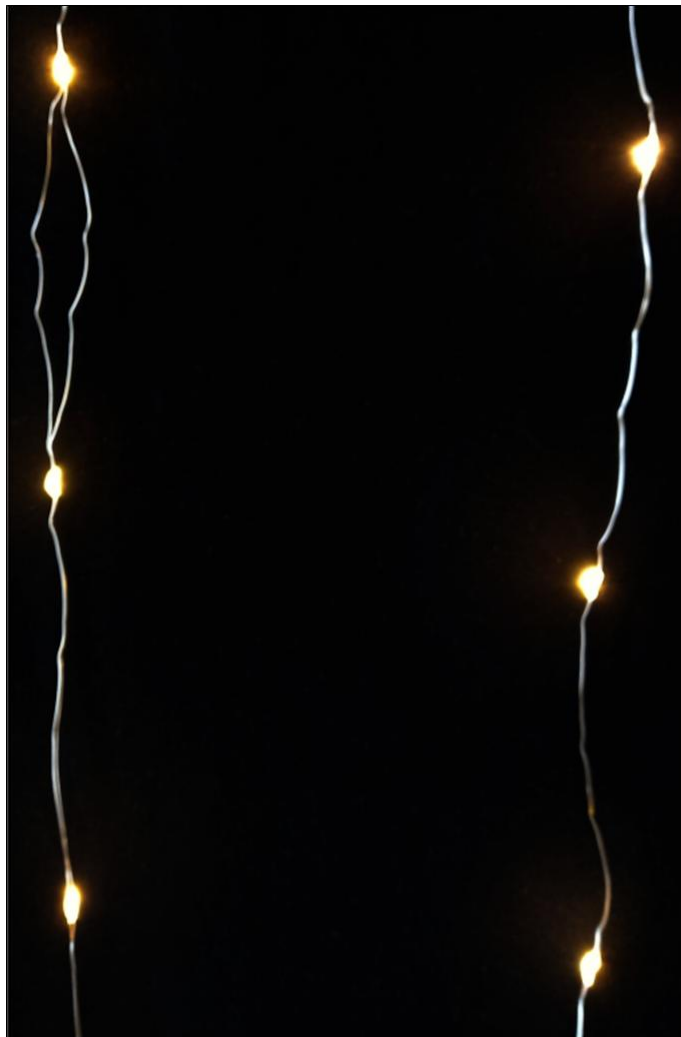
- H. Hsu: Schaum's Outline of Signals and Systems
<https://uet.vnu.edu.vn/~tantd/ss/TheoryandProblemsofSS.pdf>
- H. Babić: Signali i sustavi
http://sis.zesoi.fer.hr/predavanja/pdf/sis_2001_skripta.pdf
- B. Jeren: Signali i sustavi
<https://shop.skolskaknjiga.hr/signali-i-sustavi/>
- I.S. Pandžić et al: Uvod u teoriju informacije i kodiranje (3. poglavlje)
<https://element.hr/artikli/293/uvod-u-teoriju-informacije-i-kodiranje>

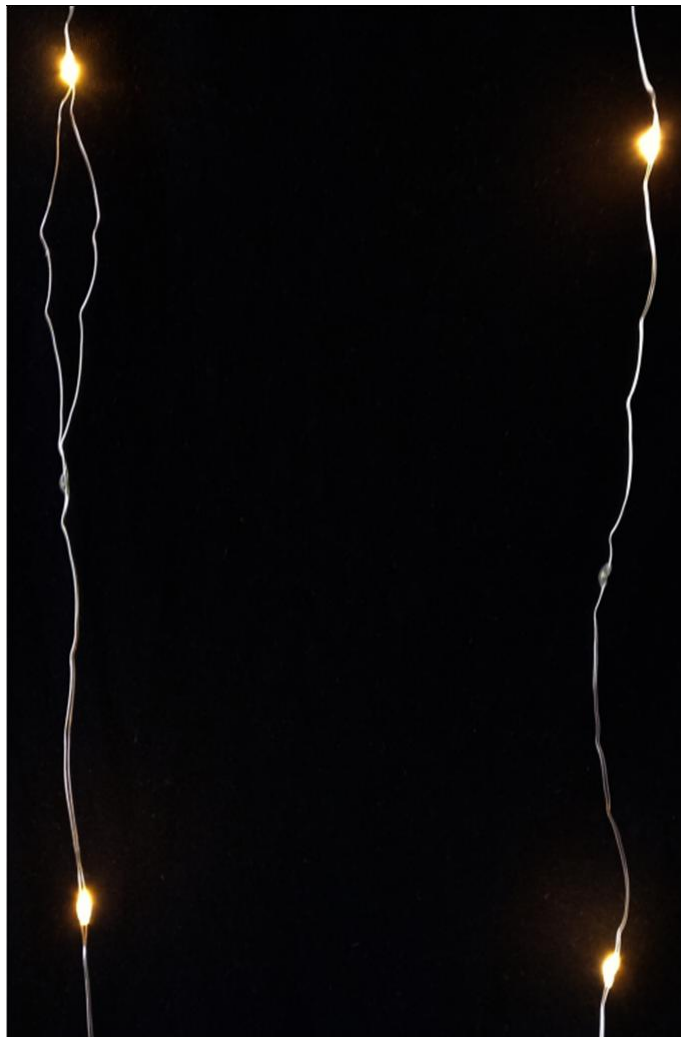
Navedena literatura je ujedno i izvor za pripremu ove prezentacije.

Promatrajte svijet oko sebe kao inženjeri...

I uvijek se zapitajte: ***Kako je neki uređaj napravljen, „kako radi” ?***





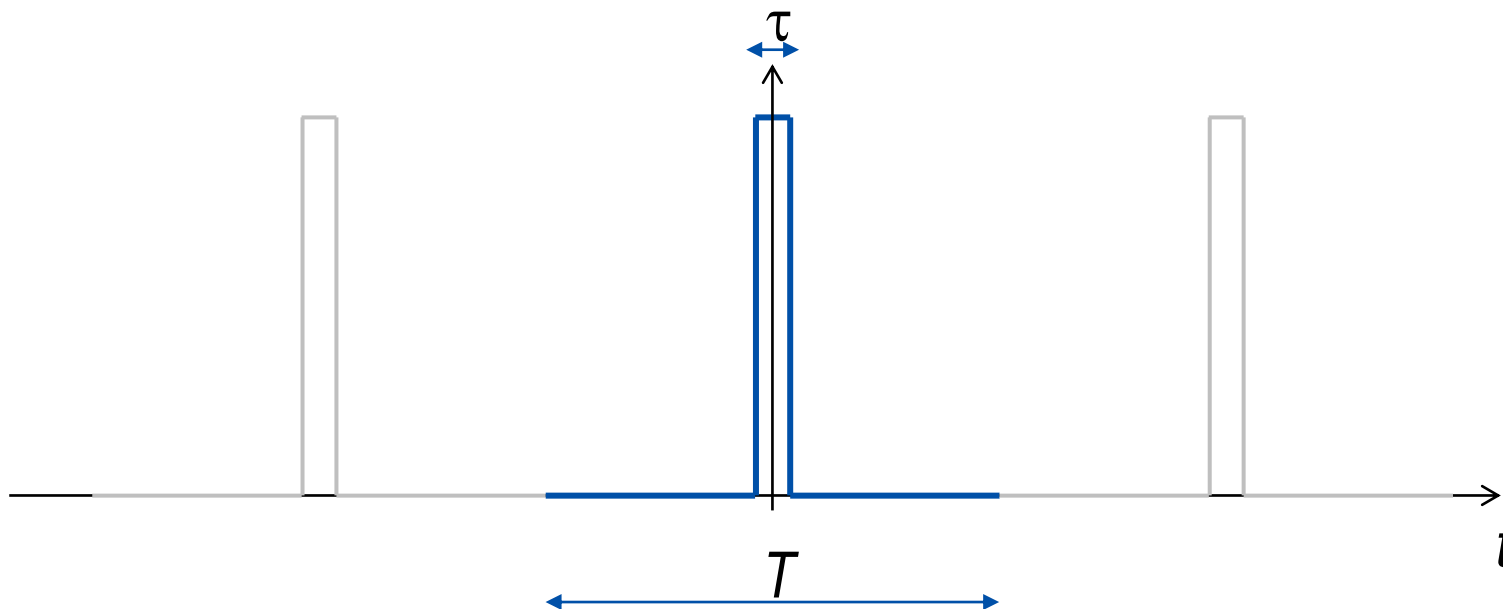




Fourierova transformacija

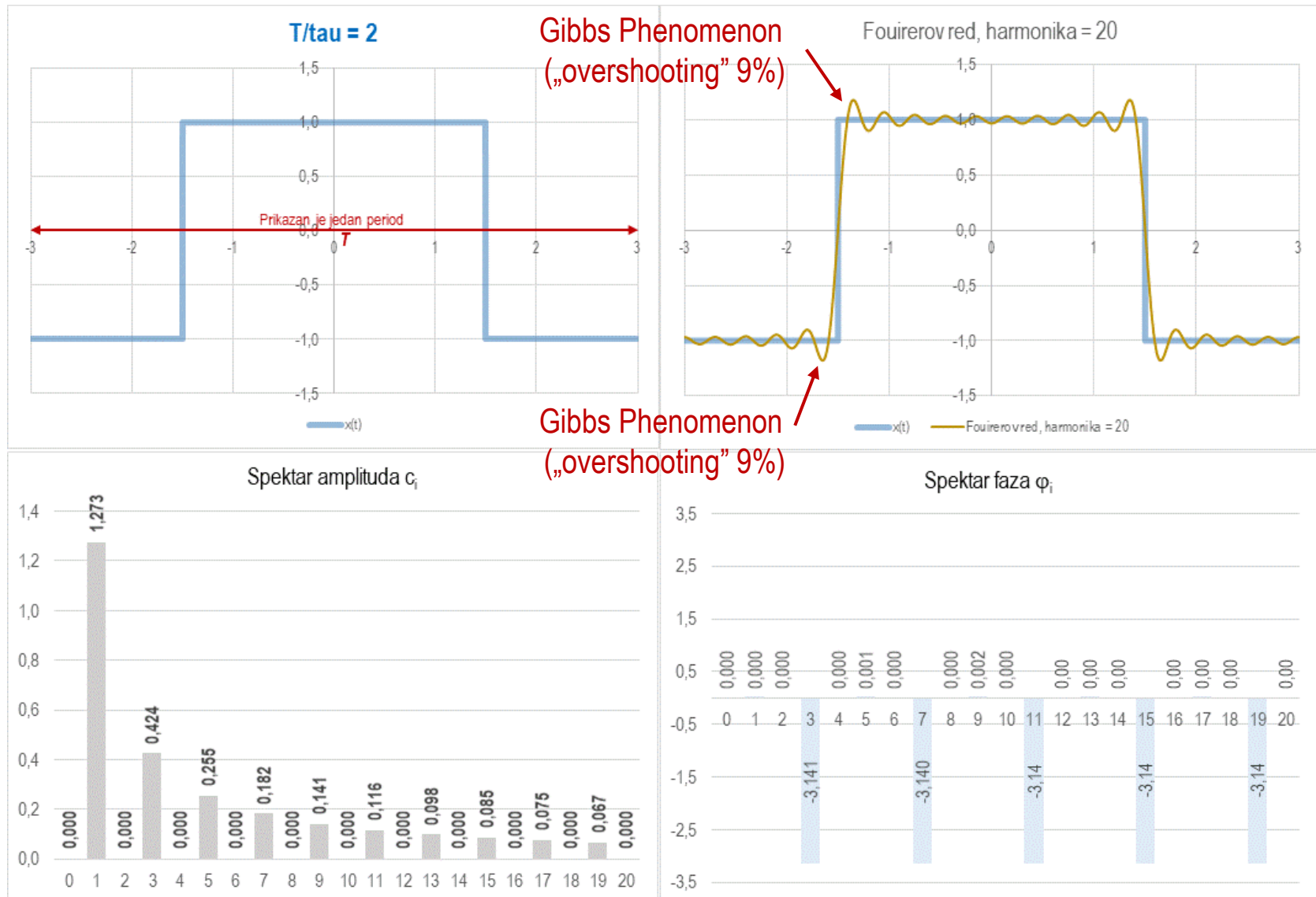
Pogledati: <https://www.youtube.com/watch?v=spUNpyF58BY>

Periodični pravokutni signal s kratkim trajanjem pozitivnog dijela impulsa τ

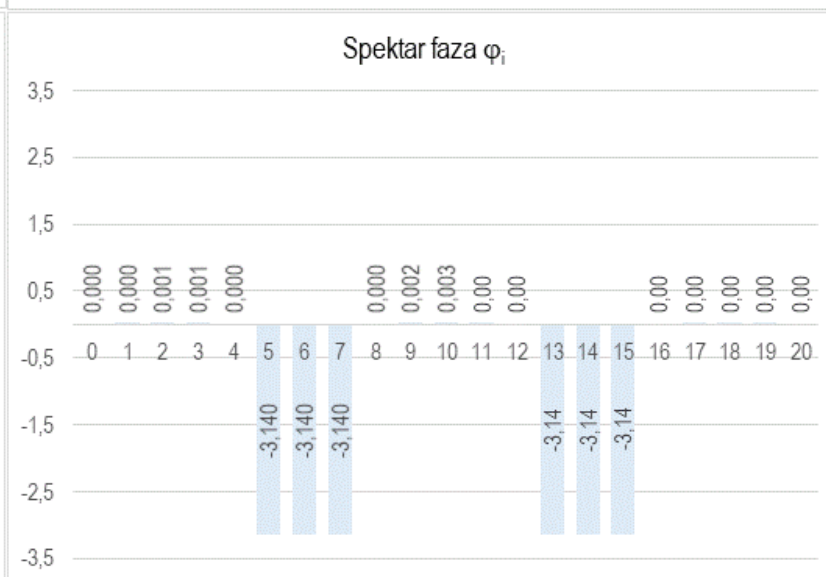
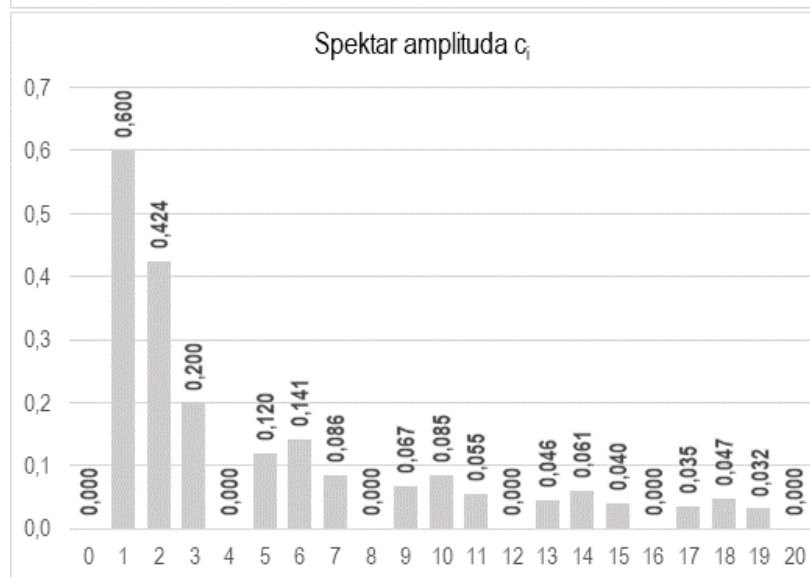
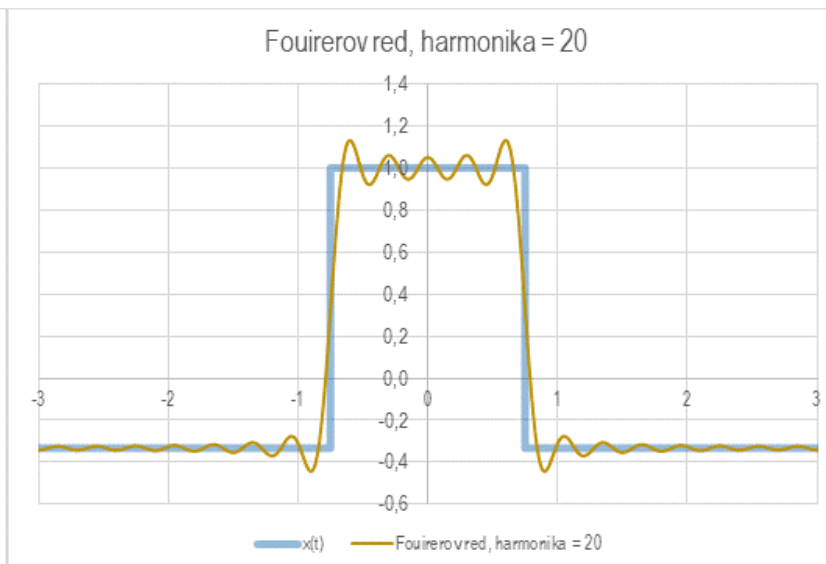
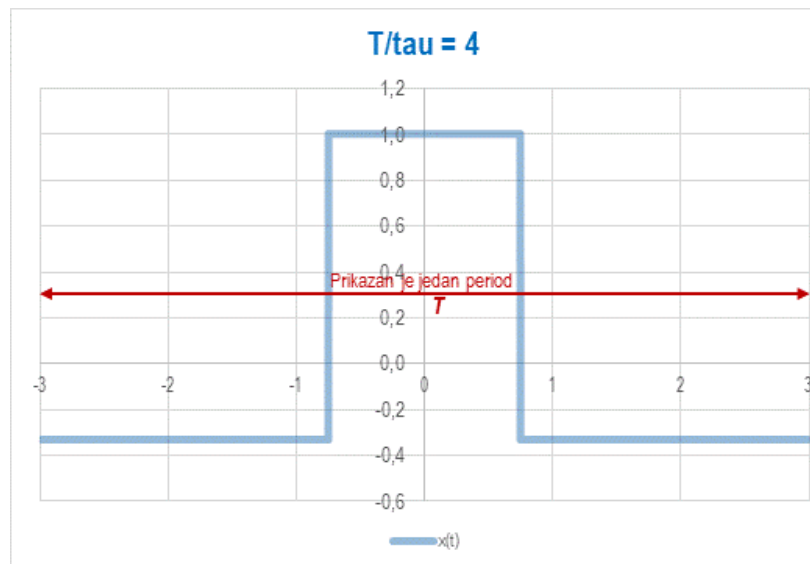


$$T / \tau = 2; 4; 8; 16; 32, 64, 128$$

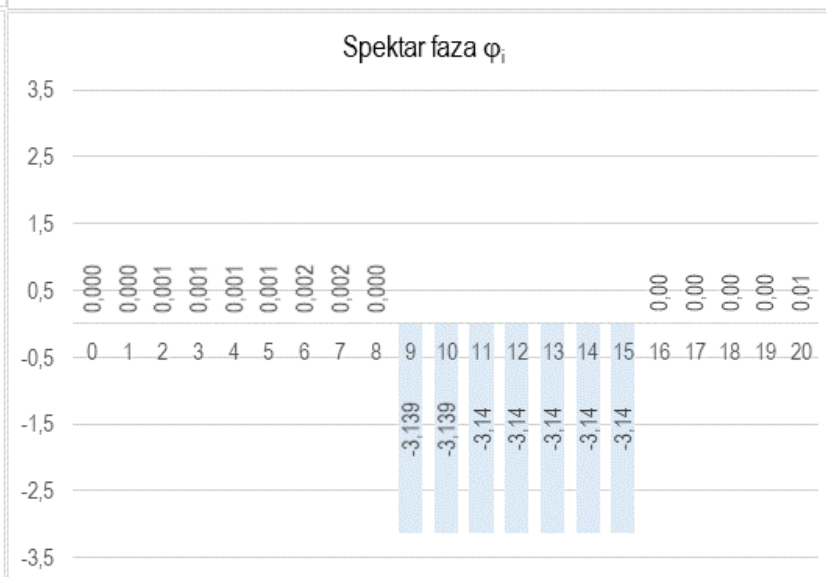
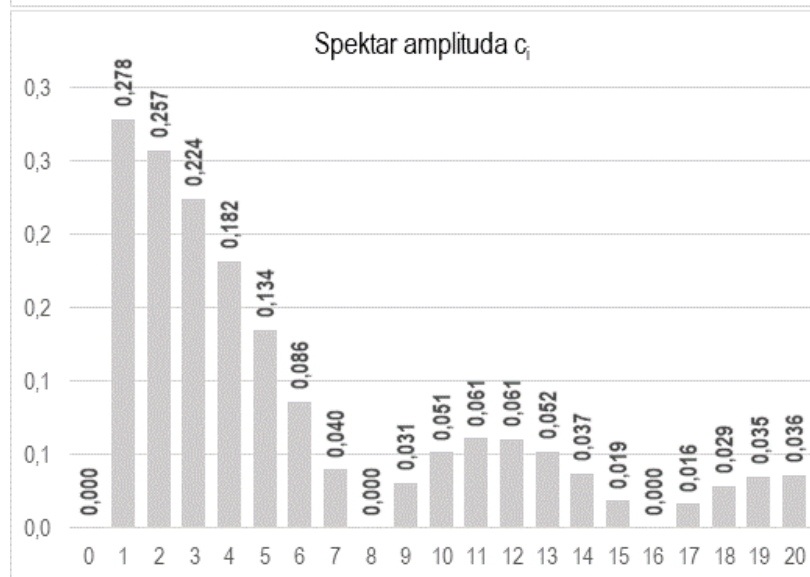
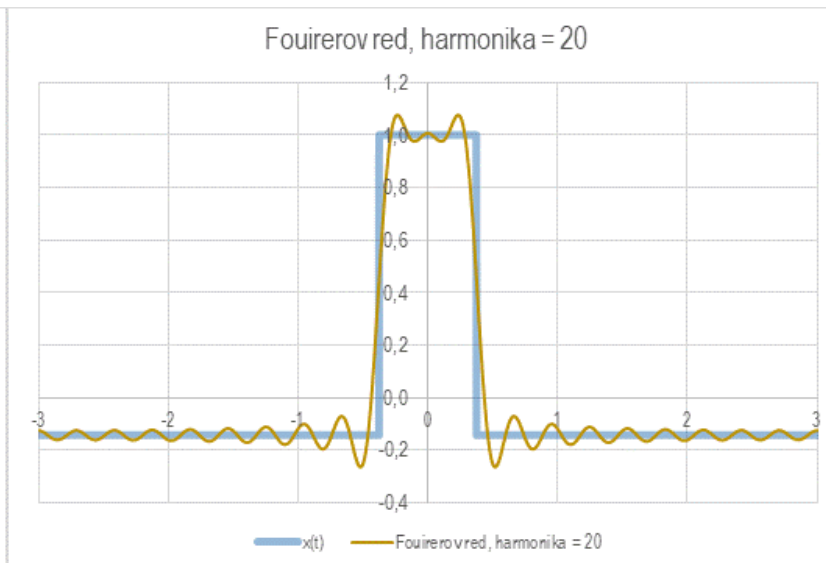
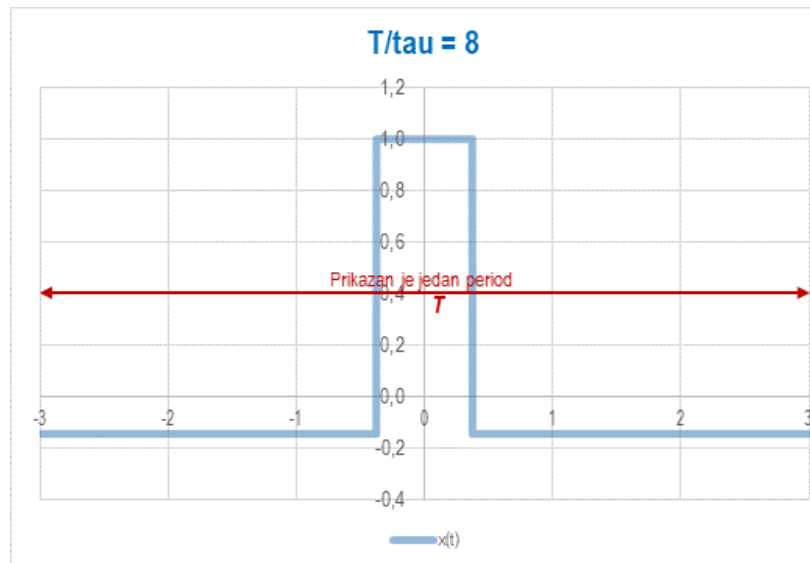
Primjer 1: $T / \tau = 2$



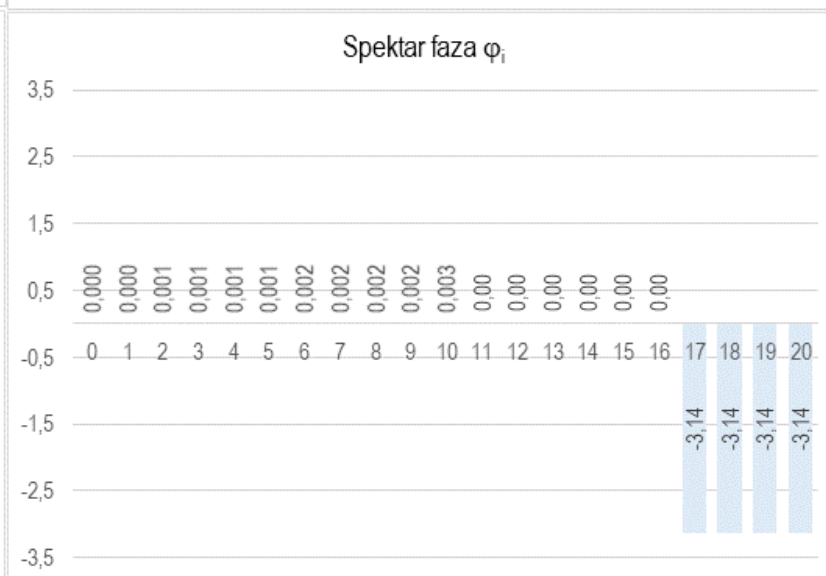
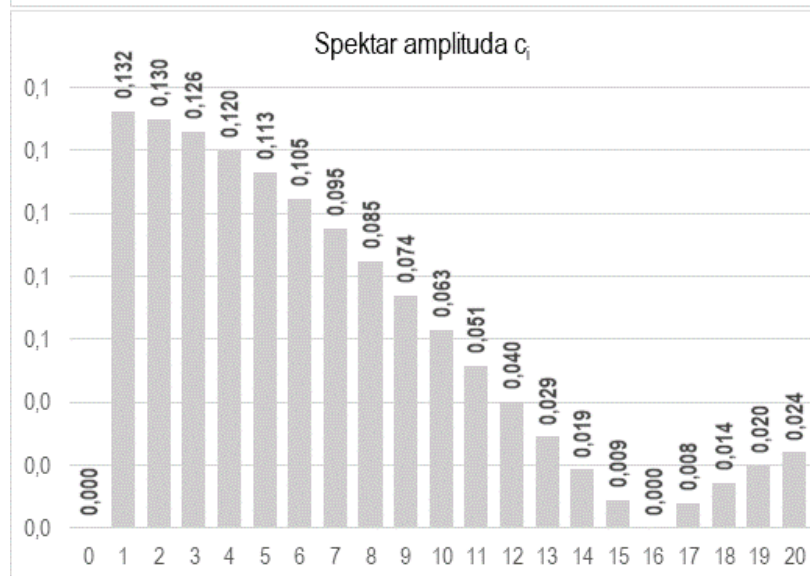
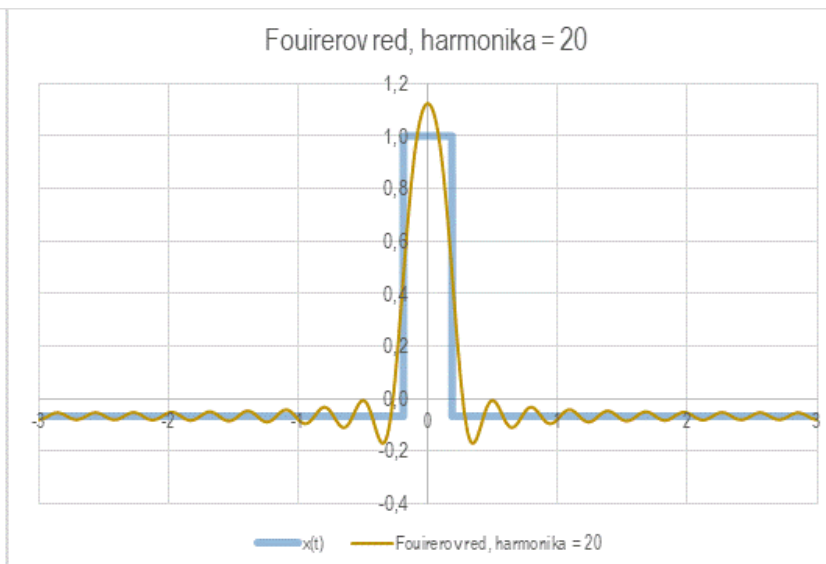
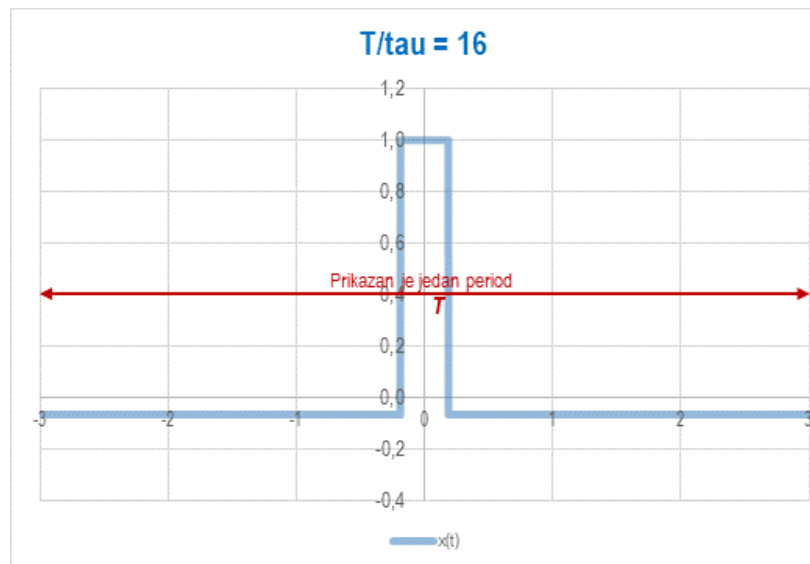
Primjer 2: $T / \tau = 4$



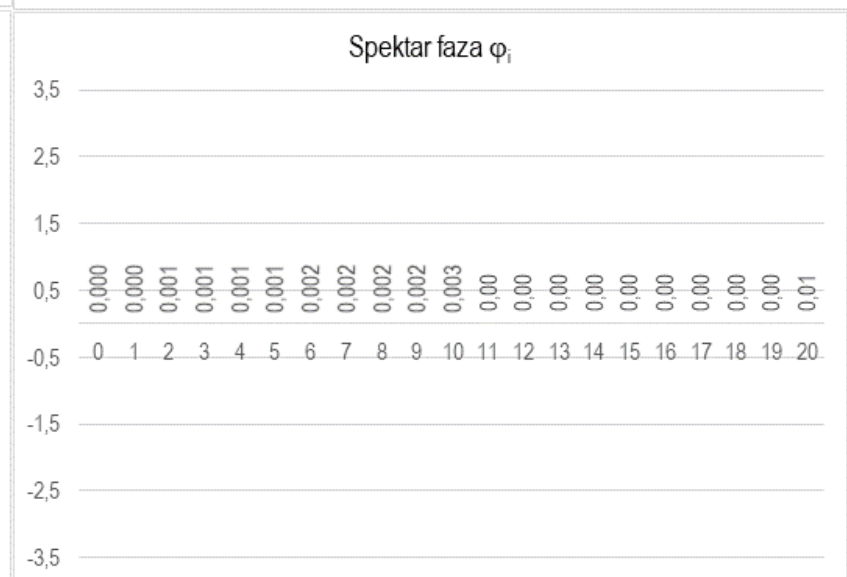
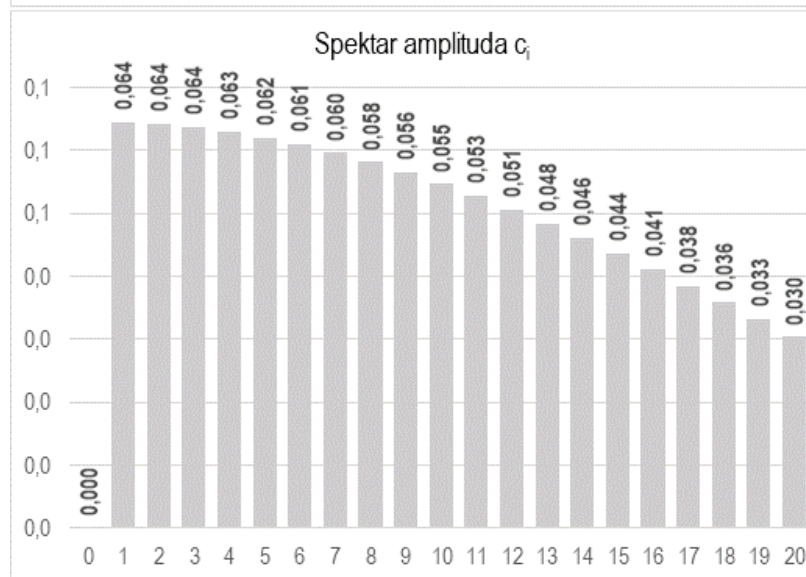
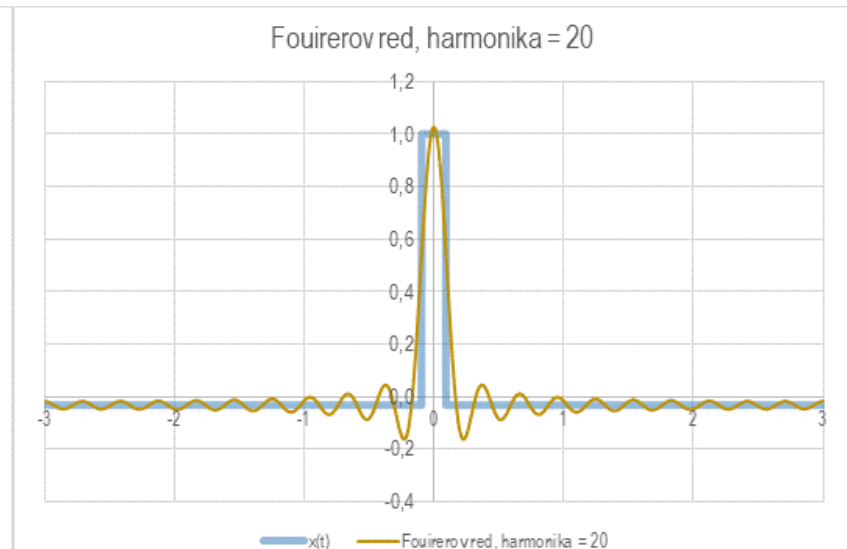
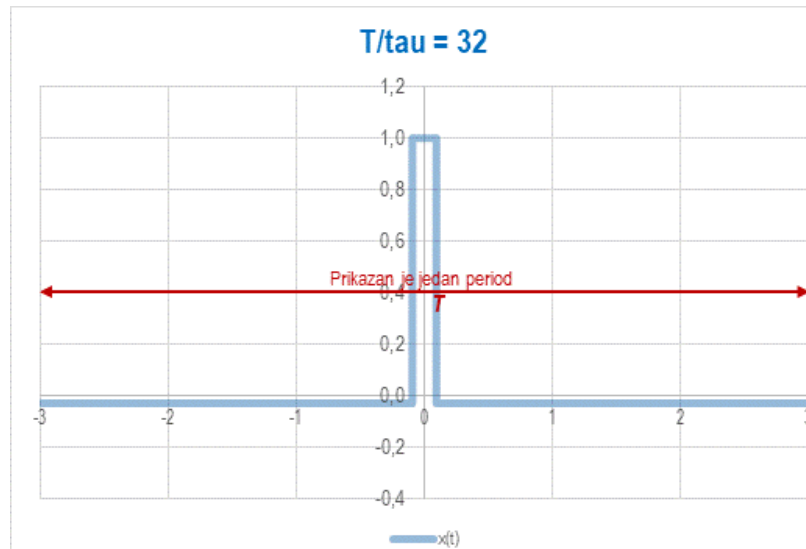
Primjer 3: $T / \tau = 8$



Primjer 4: $T / \tau = 16$



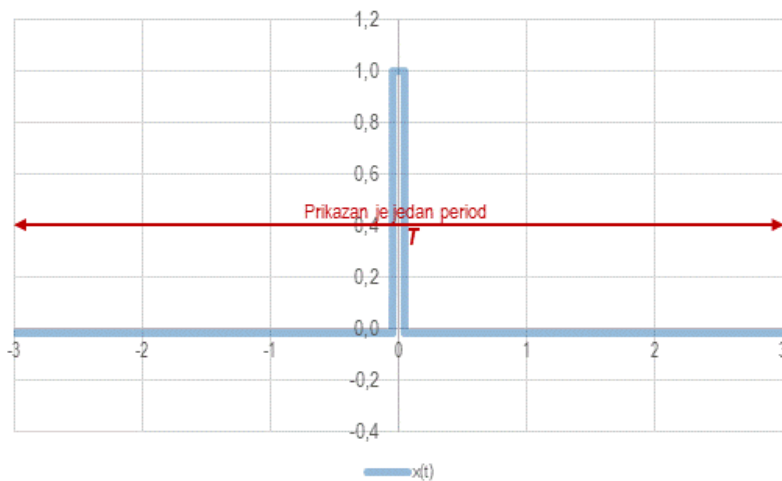
Primjer 5: $T / \tau = 32$



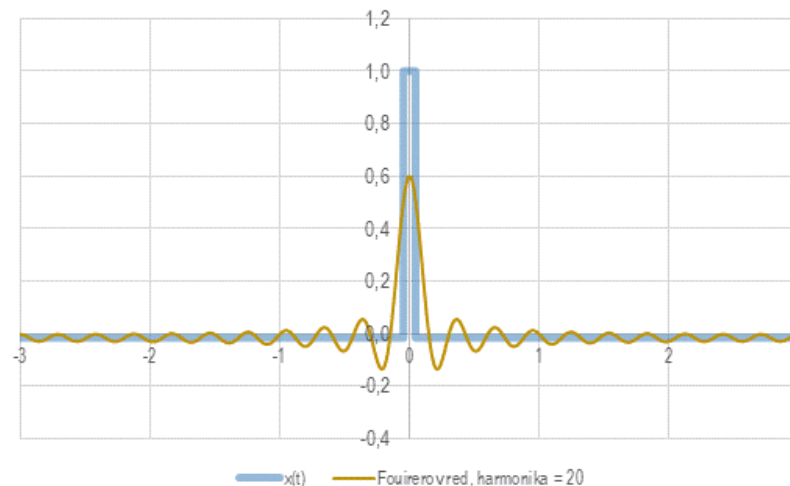
Primjer 6: $T / \tau = 64$



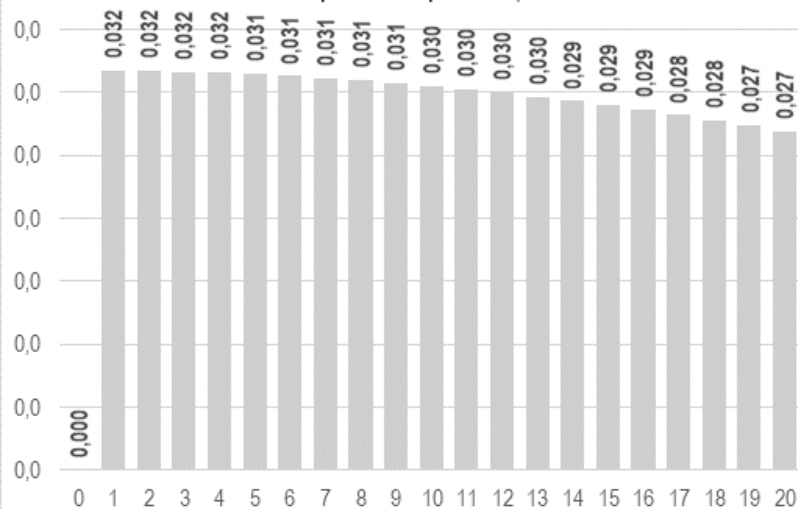
$T/\tau = 64$



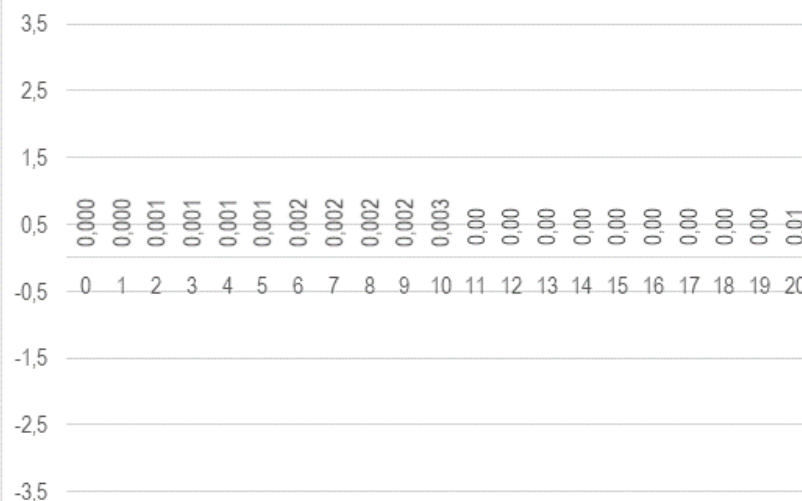
Fouirerov red, harmonika = 20



Spektar amplituda c_i



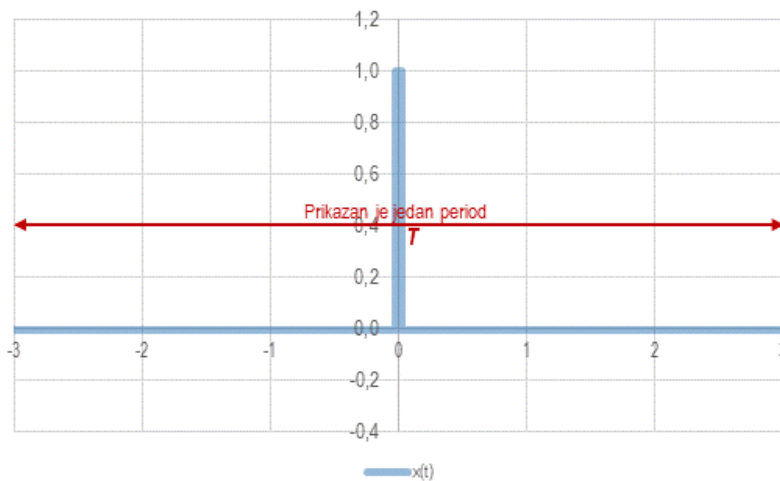
Spektar faza φ_i



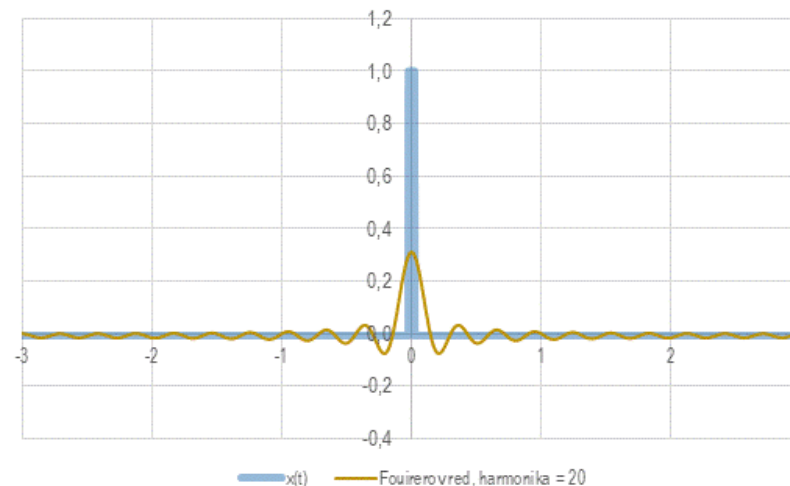
Primjer 7: $T / \tau = 128$



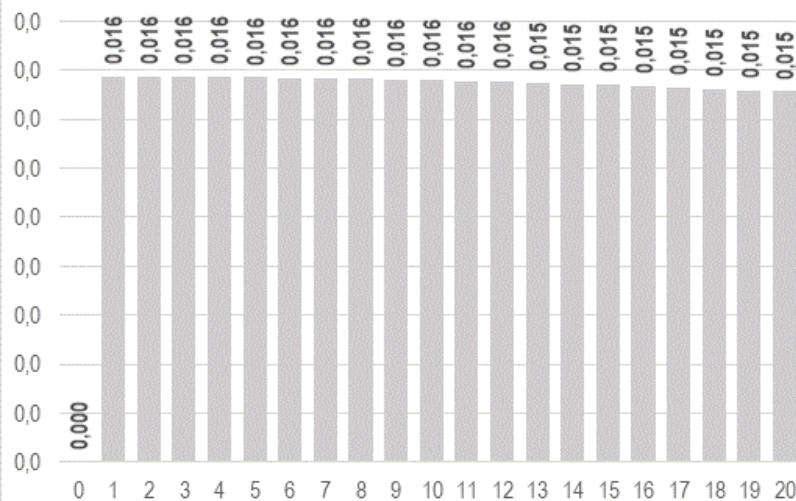
$T/\tau = 128$



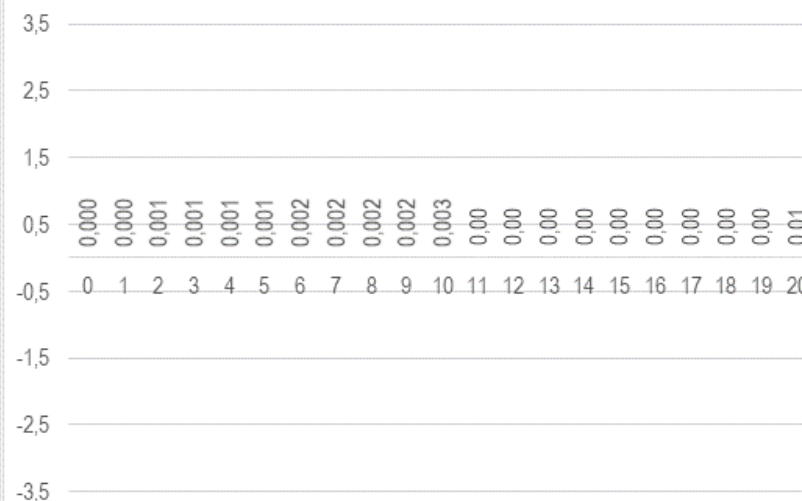
Fouirerov red, harmonika = 20



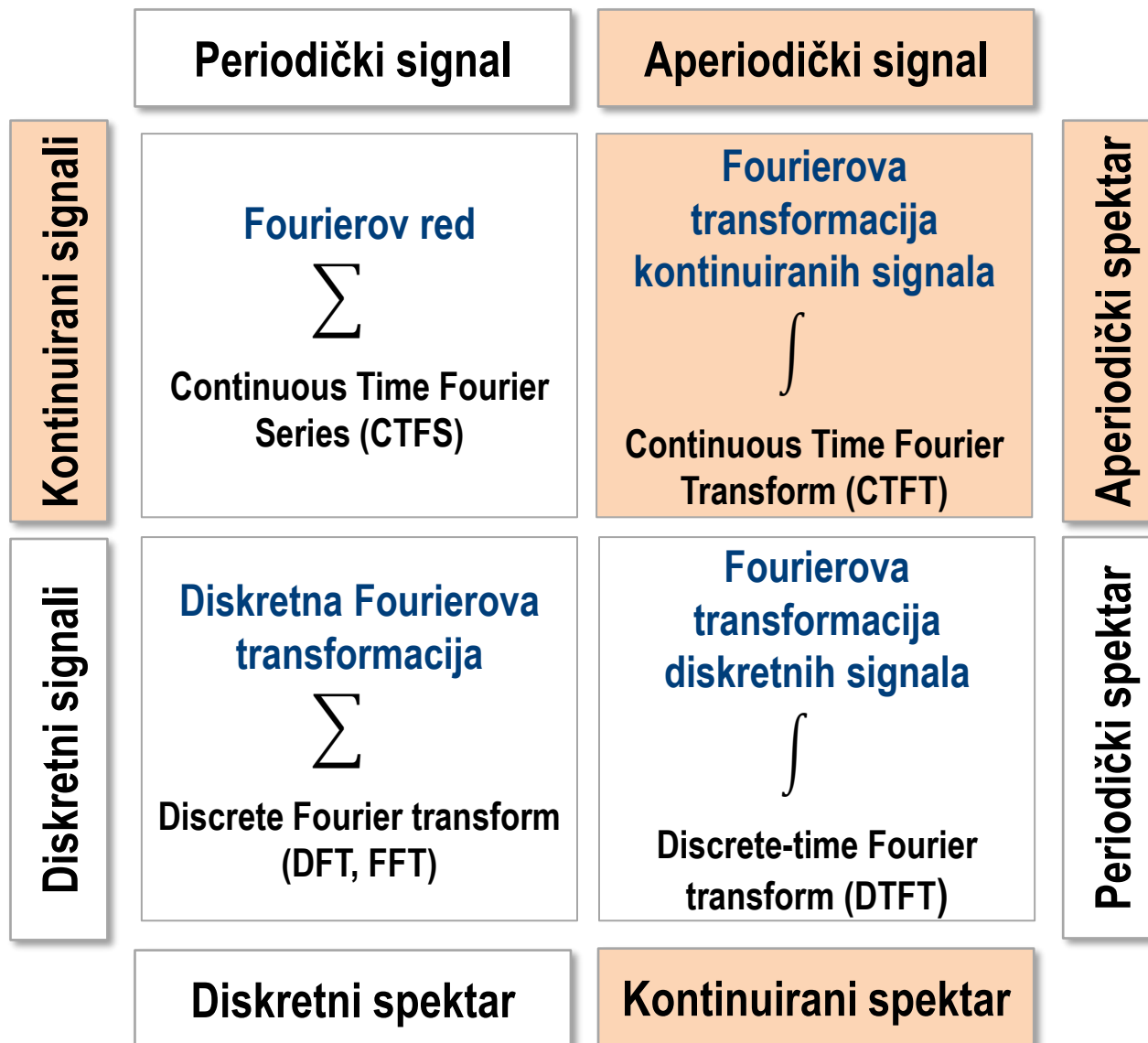
Spektar amplituda c_i



Spektar faza ϕ_i



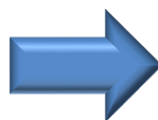
Fourierova transformacija kontinuiranih signala



Zaključci o Fourierovu redu periodičnog signala, mogu se primijeniti i na **aperiodični*** signal, tako što se period povećava do beskonačne vrijednosti.

Zbrajanje neprebrojivo mnogo diskretnih komponenti reda prerasta u integral kontinuirane funkcije spektra:

$$x(t) = \sum_{i=-\infty}^{+\infty} [c_i \cdot e^{i \cdot j\omega_1 t}]$$



$$x(t) = \int_{-\infty}^{+\infty} X(f) e^{j2\pi f t} df$$

$$X(f) = \int_{-\infty}^{+\infty} x(t) e^{-j2\pi f t} dt$$

*Aperiodičan je signal onaj koji se ponavlja “nikada”, tj. nakon beskonačno dugog vremena!

Terminologija:

 **Zapamtiti!**

direktna Fourierova
transformacija:

$$X(f) = \int_{-\infty}^{+\infty} x(t) e^{-j2\pi f t} dt$$

inverzna Fourierova
transformacija:

$$x(t) = \int_{-\infty}^{+\infty} X(f) e^{j2\pi f t} df$$

Fourierov transformacijski par

Ako je varijabla ω a ne f , izraz za $x(t)$ se množi s $1/(2\pi)$:

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} X(\omega) e^{j\omega t} d\omega$$

Formalna razlika izraza:

$$x(t) = \int_{-\infty}^{+\infty} X(f) e^{j2\pi f t} df$$

$$X(f) = \int_{-\infty}^{+\infty} x(t) e^{-j2\pi f t} dt$$

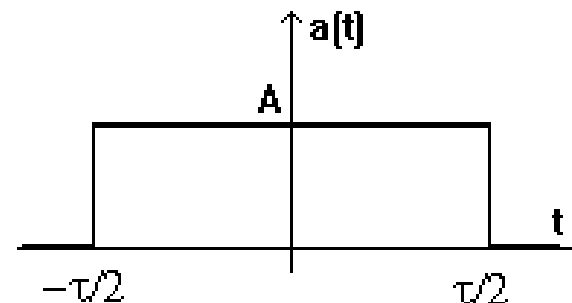
jest u **predznaku** eksponenta, što pak znači, da se zaključci o svojstvima **direktne transformacije** odnose i na **inverznu transformaciju**.

Osnovni praktični uvjet, ali ne i nužan, za postojanje Fourierove transformacije jest da je $x(t)$ apsolutno integrabilna funkcija na intervalu $\langle -\infty, +\infty \rangle$:

$$\int_{-\infty}^{+\infty} |x(t)| dt < \infty$$

Primjer: osamljeni pravokutni impuls

$$a(t) = \begin{cases} A, & |t| < T/2 \\ 0, & |t| > T/2 \end{cases}$$



$$X(f) = \int_{-\infty}^{+\infty} x(t) e^{-j2\pi f t} dt \quad \longrightarrow \quad A(\omega) = A \int_{-\tau/2}^{\tau/2} e^{-j\omega t} dt$$

$$\int e^{ax} dx = \frac{1}{a} \int e^x dx$$

$$A(\omega) = \frac{A}{-j\omega} \left[e^{-j\omega \tau/2} - e^{j\omega \tau/2} \right] \bigg/ \frac{2}{2}$$

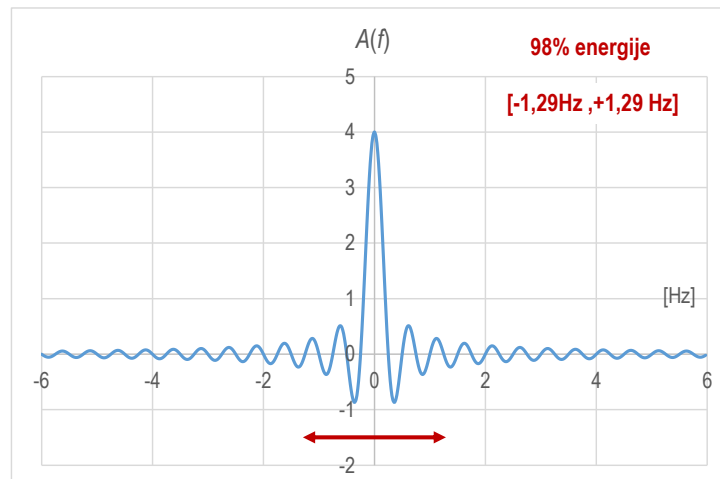
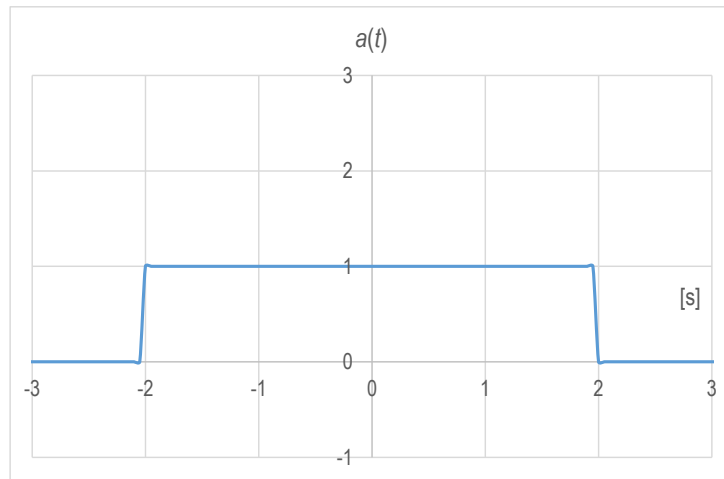
$$A(\omega) = \frac{2A}{\omega} \frac{\left[e^{j\omega \tau/2} - e^{-j\omega \tau/2} \right]}{2j}$$

$$A(f) = A \frac{\sin(\pi f \tau)}{\pi f}$$

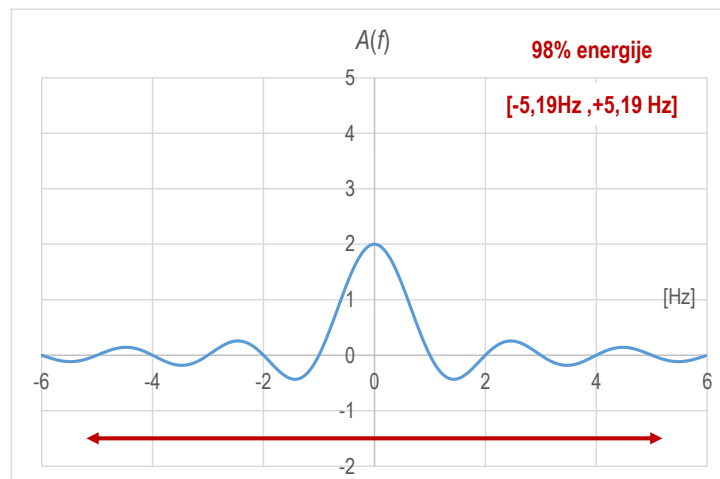
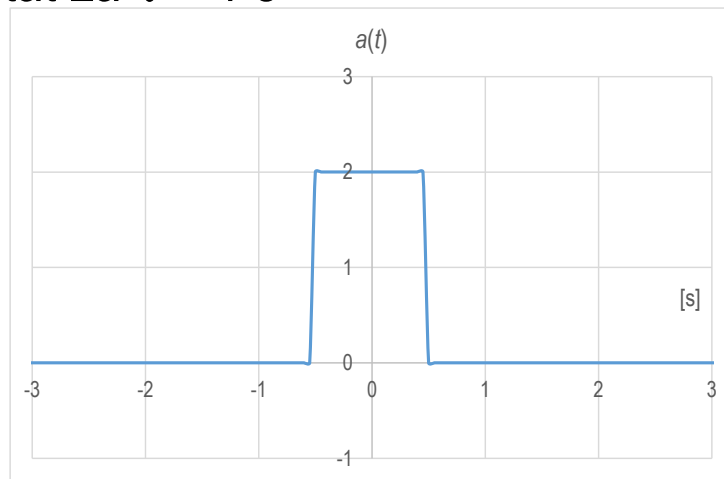
Fourierova transformacija: Impulsi iste energije



Rezultat za $\tau = 4$ s



Rezultat za $\tau = 1$ s



Zaključak?

Parsevalov teorem



Parsevalov teorem za Fourerov red u kompleksnom obliku:

$$\sum_{i=-\infty}^{+\infty} |c_i|^2 = \frac{1}{T} \int_T u^2(t) dt$$

$$P_{sr} = \frac{1}{R} \sum_{i=-\infty}^{+\infty} |c_i|^2 = \frac{1}{RT} \int_T u^2(t) dt = \frac{U_{ef}^2}{R}$$

Parsevalov teorem za Fourierovu transformaciju kontinuiranih signala:

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df$$



Zzv14-01 Provjerite ispravnost Parsevalovog teorem za Fourierovu transformaciju na primjeru osamljenog pravokutnog impulsa.



Linearnost transformata:

Originalni signali $x(t)$ i $y(t)$ imaju Fourierove transformate $X(f)$ i $Y(f)$. Linearnoj kombinaciji signala $z(t) = a \cdot x(t) + b \cdot y(t)$, pripada transformat $Z(f)$ jednak linearnoj kombinaciji transformata pojedinačnih signala:

$$z(t) = a x(t) + b y(t) \rightarrow Z(f) = a X(f) + b Y(f)$$

Simetrija transformata (transformat od transformata):

$$x(t) \rightarrow X(f) \qquad X(t) = x(-f)$$

Promjena nezavisne varijable:

$$x(t) \rightarrow X(f) \qquad x(at) \rightarrow \frac{1}{|a|} X(f/a)$$

Svojstva Fourierove transformacije (2/3)



Vremenski pomak signala:

Vremenski pomak signala $x(t)$ za t_0 , manifestira se promjenom **samo** u **faznom** dijelu transformata:

$$x(t) \rightarrow X(f) \qquad x(t-t_0) \rightarrow e^{-j\omega t_0} X(f)$$

Frekvencijski pomak spektra:

Pomaku transformata u frekvencijskoj domeni za f_0 , u vremenskoj domeni odgovara množenju signala s $e^{j\omega_0 t}$

$$x(t) \rightarrow X(f) \qquad e^{j\omega_0 t} x(t) \rightarrow X(f-f_0)$$

Deriviranje po vremenu:

$$x(t) \rightarrow X(f) \qquad \frac{d^n x(t)}{dt^n} \rightarrow (j f)^n X(f)$$

Deriviranje po frekvenciji:

$$x(t) \rightarrow X(f) \qquad (-j t)^n x(t) \rightarrow \frac{d^n X(f)}{df^n}$$

Svojstva Fourierove transformacije (3/3)



Konvolucija signala $x(t)$ i $y(t)$ u vremenskoj domeni definirana je integralom:

$$z(\tau) = \int_{-\infty}^{\infty} x(t) y(\tau - t) dt$$

Uobičajena oznaka za konvoluciju je $\mathbf{x(t) * y(t)}$.

Ako je $x(t) \rightarrow X(f)$ i $y(t) \rightarrow Y(f)$ tada **konvoluciji** funkcija $x(t)$ i $y(t)$ u vremenskoj domeni, odgovara **produkt** Fourierovih transformata u frekvencijskoj domeni:

$$\int_{-\infty}^{\infty} x(t) y(\tau - t) dt \rightarrow X(f) Y(f)$$

Slično vrijedi i za konvoluciju transformata u frekvencijskoj domeni:

$$x(t) y(t) \rightarrow \int_{-\infty}^{\infty} X(f_1) Y(f - f_1) df_1$$

Parsevalova formula:

$$x(t) \rightarrow X(f) \qquad \int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df$$